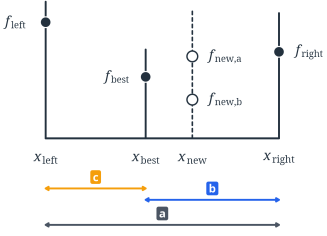
Optimization in Machine Learning

Univariate optimization
Golden section search

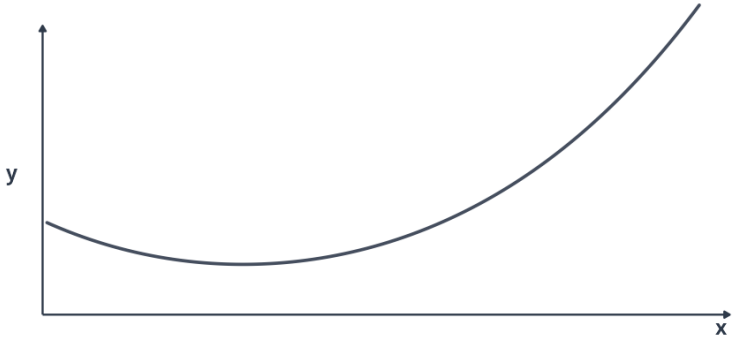


Learning goals

- Simple nesting procedure
- Golden section search

UNIVARIATE OPTIMIZATION

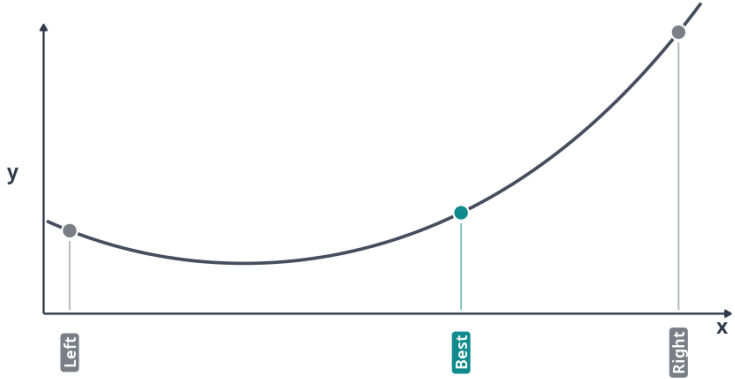
- Setting: optimize univariate $f : \mathbb{R} \to \mathbb{R}$ over interval $(x_{\text{left}}, x_{\text{right}})$
- Assumption: $-f$ is unimodal and interval $(x_{\text{left}}, x_{\text{right}})$ (called a **bracket**) contains the minimum
- Goal: iteratively shrink the bracket (**bracketing**); gradient-free, works for black-box f



SIMPLE NESTING PROCEDURE

Let $f : \mathbb{R} \to \mathbb{R}$

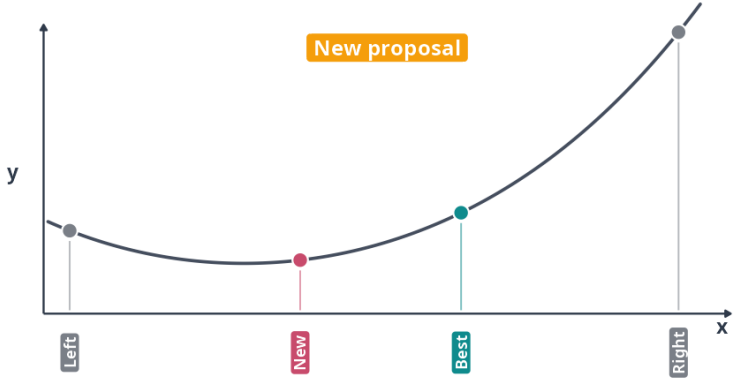
Always maintain three points: left, right, and current best



SIMPLE NESTING PROCEDURE

Let $f : \mathbb{R} \to \mathbb{R}$

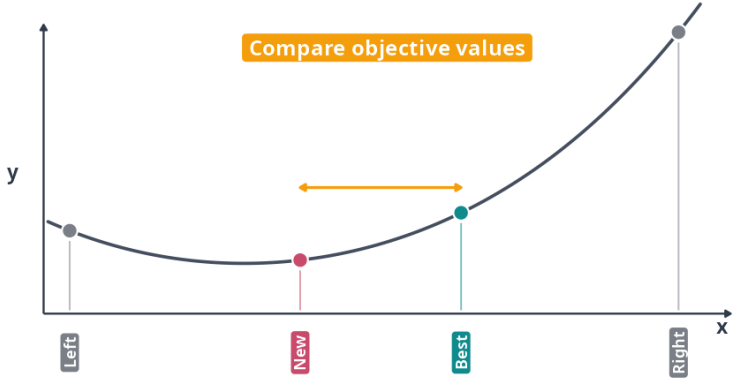
Propose random interval point (def. optimal new proposal later)



SIMPLE NESTING PROCEDURE

Let $f : \mathbb{R} \to \mathbb{R}$

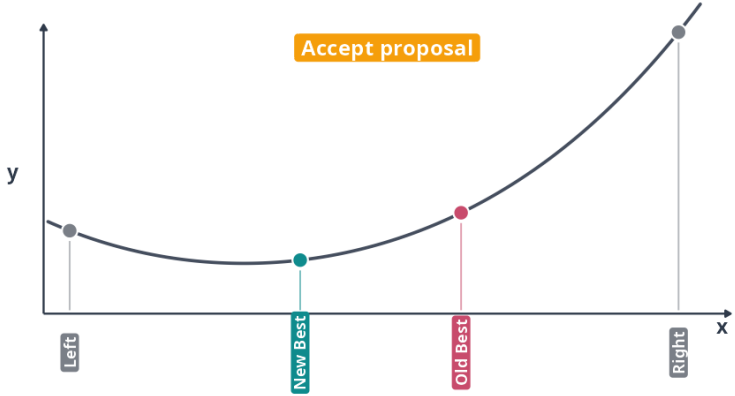
Compare proposal against current best



SIMPLE NESTING PROCEDURE

Let $f : \mathbb{R} \to \mathbb{R}$

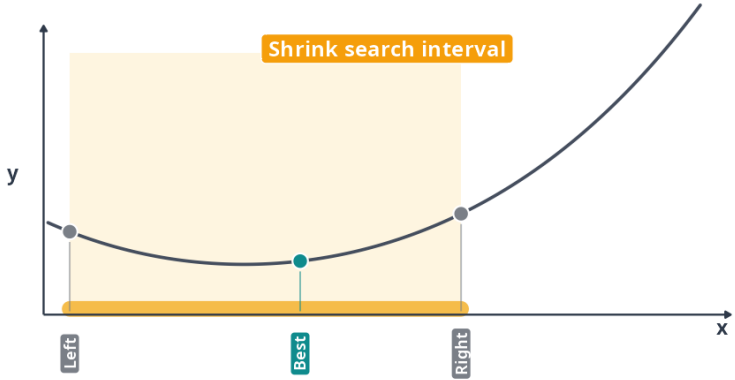
If it is better: proposal becomes current best



SIMPLE NESTING PROCEDURE

Let $f : \mathbb{R} \to \mathbb{R}$

New search interval: around current best



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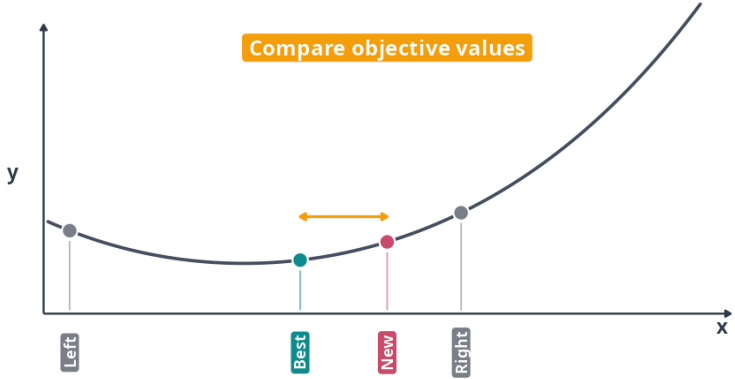

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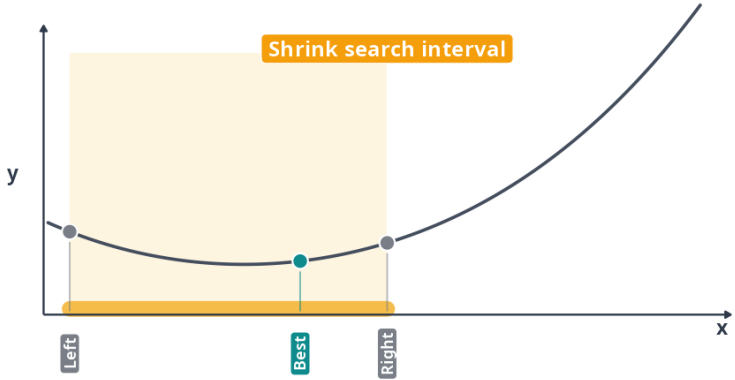

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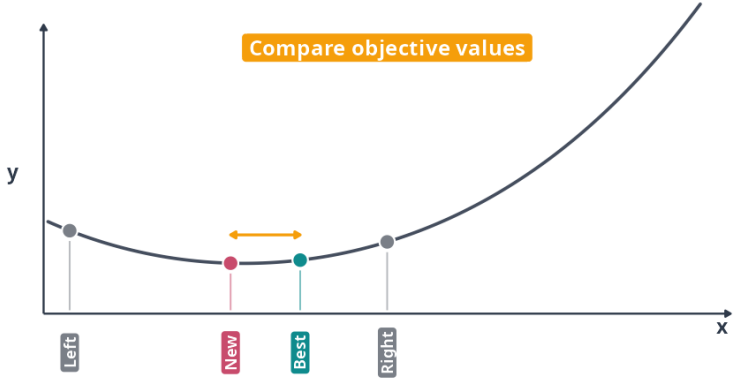

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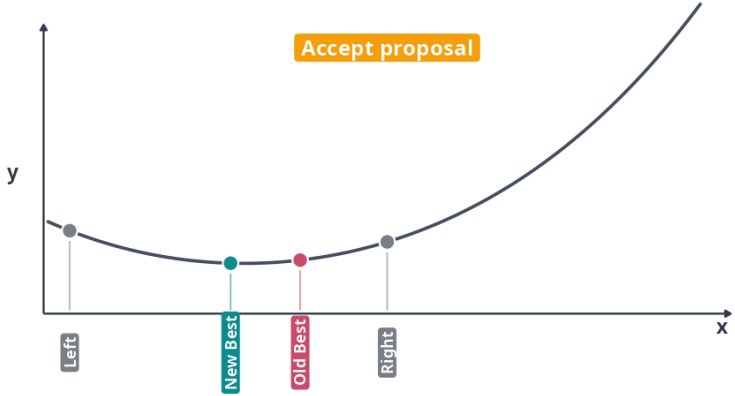

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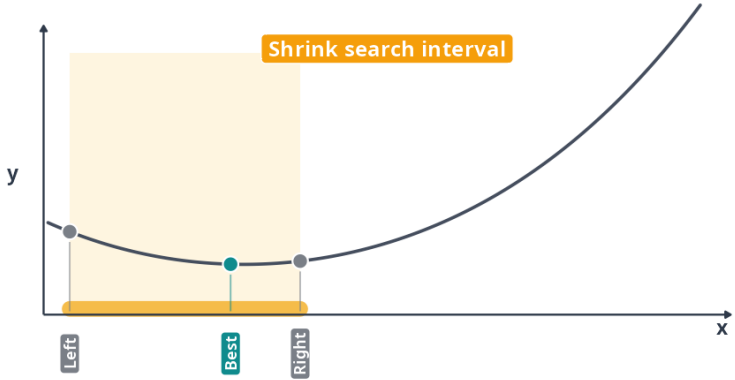
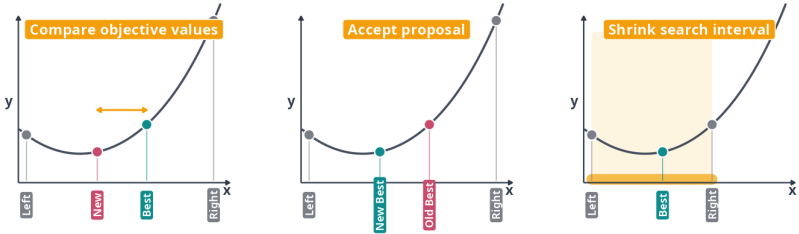


SIMPLE NESTING PROCEDURE

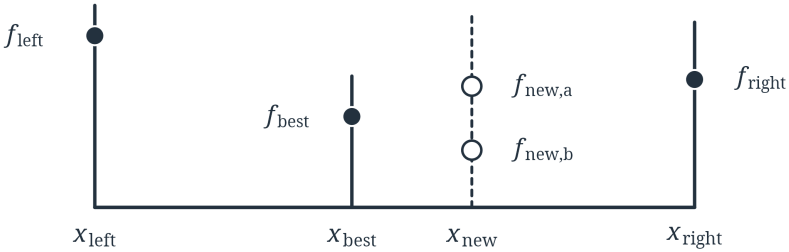
- Initialization: Search interval $(x_{\text{left}}, x_{\text{right}})$, $x_{\text{left}} < x_{\text{right}}$
- Choose x_{best} randomly
- For $t = 0, 1, 2, \ldots$
  - Choose x_{new} randomly in $[x_{\text{left}}, x_{\text{right}}]$
  - If $f(x_{\text{new}}) < f(x_{\text{best}})$:
    - $x_{\text{best}} \leftarrow x_{\text{new}}$
  - New interval: points around x_{best}



GOLDEN SECTION SEARCH

Key question: How can x_{new} be chosen better than randomly?

- Principle 1: Always in bigger subinterval to maximize reduction
- Principle 2: x_{new} symmetrically to x_{best} for uniform reduction

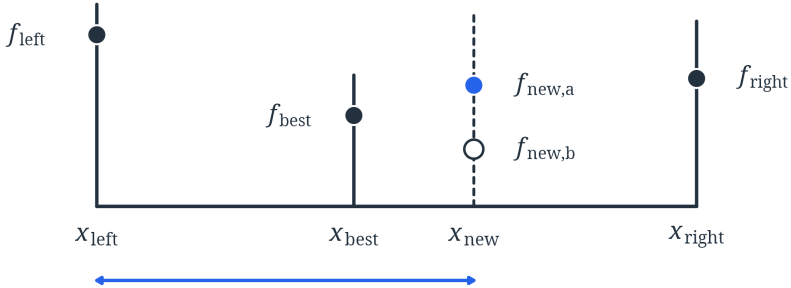


Consider two hypothetical outcomes for x_{new}: $f_{new,a}$ and $f_{new,b}$

GOLDEN SECTION SEARCH

If outcome is $f_{new,a}$, x_{best} remains best and we search around x_{best}:

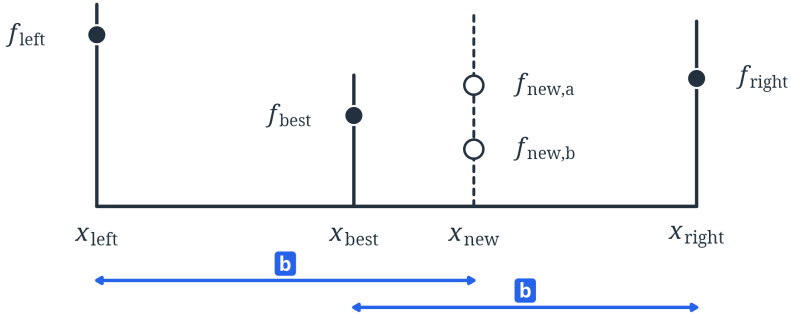
$$[x_{\text{left}}, x_{\text{new}}]$$



GOLDEN SECTION SEARCH

If $f_{new,b}$ is outcome, x_{new} becomes best point and search around x_{new}:

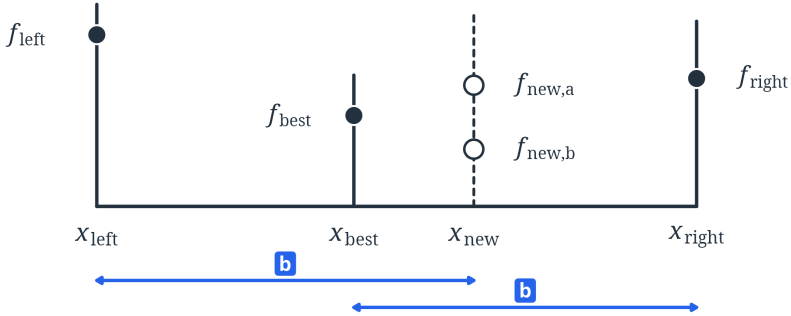
$$[x_{\text{best}}, x_{\text{right}}]$$



GOLDEN SECTION SEARCH

For uniform reduction the two potential intervals must be equal-sized:

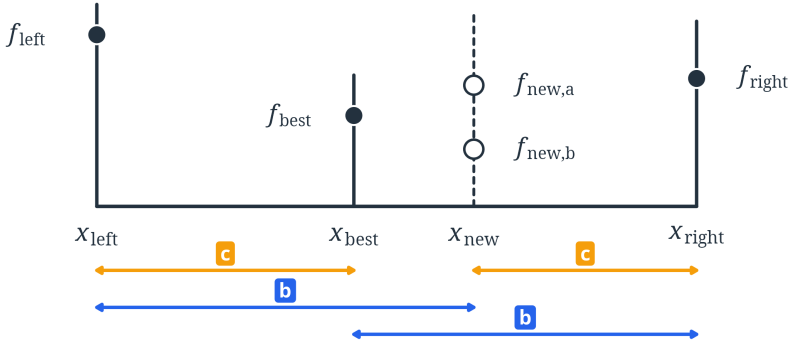
$$b := x_{\text{right}} - x_{\text{best}} = x_{\text{new}} - x_{\text{left}}$$



GOLDEN SECTION SEARCH

One iteration ahead: require again the intervals to be of same size.

$$c := x_{\text{best}} - x_{\text{left}} = x_{\text{right}} - x_{\text{new}}$$



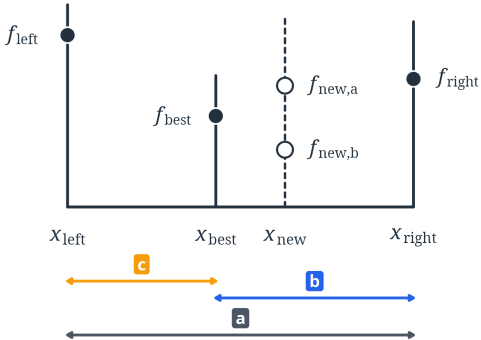
GOLDEN SECTION SEARCH

To summarize, we require:

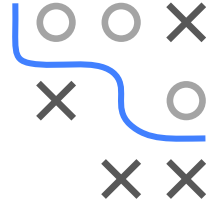
$$a = x_{\text{right}} - x_{\text{left}}$$

$$b = x_{\text{right}} - x_{\text{best}} = x_{\text{new}} - x_{\text{left}}$$

$$c = x_{\text{best}} - x_{\text{left}} = x_{\text{right}} - x_{\text{new}}$$



GOLDEN SECTION SEARCH



- We require same percentage improvement in each iteration
- For φ reduction factor of interval sizes (a to b, and b to c)

$$\varphi := \frac{b}{a} = \frac{c}{b}$$

$$\varphi^2 = \frac{b}{a} \cdot \frac{c}{b} = \frac{c}{a}$$

- Divide $a = b + c$ by a:

$$\begin{aligned} \frac{a}{a} &= \frac{b}{a} + \frac{c}{a} \\ 1 &= \varphi + \varphi^2 \\ 0 &= \varphi^2 + \varphi - 1 \end{aligned}$$

- Unique positive solution is $\varphi = \frac{\sqrt{5}-1}{2} \approx 0.618$

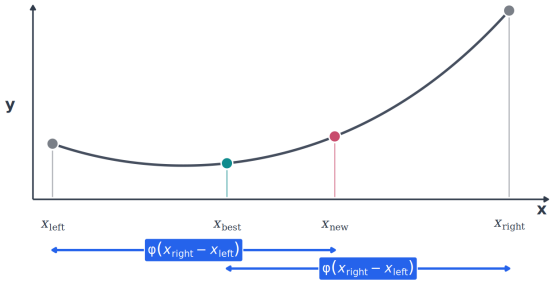
GOLDEN SECTION SEARCH

- Place x_{best} and x_{new} at $\varphi \approx 0.618$ from each endpoint:

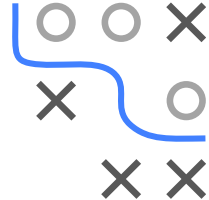
$$\begin{aligned} x_{\text{best}} &= x_{\text{right}} - \varphi(x_{\text{right}} - x_{\text{left}}) \\ &= x_{\text{left}} + (1-\varphi)(x_{\text{right}} - x_{\text{left}}) \end{aligned}$$

- By symmetry:

$$\begin{aligned} x_{\text{new}} &= x_{\text{left}} + \varphi(x_{\text{right}} - x_{\text{left}}) \\ &= x_{\text{right}} - (1-\varphi)(x_{\text{right}} - x_{\text{left}}) \end{aligned}$$



GOLDEN SECTION SEARCH — TERMINATION



- Simple criterion: stop when bracket width falls below τ:

$$|x_{\text{right}} - x_{\text{left}}| < \tau$$

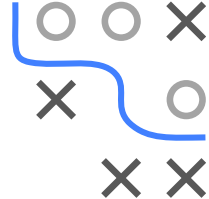
- Issue: τ is absolute — too loose near a minimum close to 0, too tight for a minimum far from 0

- Scale-invariant alternative ▸ Teukolsky et al. 2017:

$$|x_{\text{right}} - x_{\text{left}}| \leq \tau(|x_{\text{best}}| + |x_{\text{new}}|)$$

  where $|x_{\text{best}}| + |x_{\text{new}}|$ serves as a proxy for the scale of the minimum, stopping when the bracket is small relative to the minimum's location

GOLDEN SECTION SEARCH — CONVERGENCE



- Safe method: always converges (given $-f$ is unimodal on $(x_{\text{left}}, x_{\text{right}})$)
- (Slow) linear convergence: bracket shrinks by factor $\varphi \approx 0.618$ per iteration